

Semi-Automatic Generation of Drivable Road Networks from Satellite Imagery for CARLA

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Abstract—We present a lightweight SAM-to-OpenDRIVE pipeline for generating simulator-ready road networks for CARLA from overhead imagery. Road masks obtained via prompt-based segmentation are converted into vector graphs and compiled into geometrically consistent ASAM OpenDRIVE maps with valid lanes and junction topology. The modular design enables scalable and rapid map prototyping while explicitly targeting simulator ingestion and structural compliance.

Keywords—*Index Terms*—Road network extraction; satellite imagery; computer vision; Segment Anything Model; OpenDRIVE; CARLA simulator; map generation; autonomous driving simulation; traffic modeling

I. INTRODUCTION

High-definition (HD) maps have become a de-facto enabling asset for autonomous driving research and validation, because they provide a structured prior over road geometry, topology, and traffic rules that downstream modules can consume for localization, route planning, and scenario generation. In parallel, simulation has matured into a core instrument for scaling development and testing, allowing teams to iterate on perception and planning stacks under controllable conditions before deploying to the real world. CARLA is a widely adopted open-source simulator that supports reproducible experiments, configurable sensors, and a growing ecosystem of benchmarks and tooling [1]. In practice, however, the map layer remains a critical bottleneck: building realistic road networks with correct junction topology and simulator-compatible semantics is still largely a manual, time-intensive engineering activity.

A large part of the current simulation ecosystem relies on standardized road network representations, most notably ASAM OpenDRIVE, which encodes a static road network (roads, lanes, junctions, traffic elements, and parametric geometries) in an exchangeable XML-based format [2]. OpenDRIVE has strong traction because it provides a common denominator across many traffic and driving simulators, and it is specifically designed to represent smooth lane-level road geometries and intersection structures. At the same time, the OpenDRIVE data model is strict: it expects geometrically consistent reference lines, continuous curvature profiles, explicit junction definitions, and coherent lane connectivity. Empirical studies on validating parametric OpenDRIVE road space models highlight how easily inconsistencies can arise

when the underlying geometry is noisy or when constraints (e.g., curvature continuity, lane boundary correctness) are not systematically enforced [3]. Consequently, the “last mile” from a crude road centerline estimate to a simulator-ready OpenDRIVE map can dominate the total effort and frequently requires expert-level map editing.

This motivates the following strategic direction: automate the map authoring pipeline by extracting road networks from overhead imagery (satellite or aerial photos), converting them into a topologically correct vector road graph, and then compiling that graph into an OpenDRIVE-compliant representation that can be loaded into CARLA and related simulators. Overhead imagery is attractive because it is globally available, scalable, and often updated more frequently than authoritative HD map products. From an engineering standpoint, such a pipeline would unlock a significantly faster content lifecycle for simulation: “map anywhere” becomes feasible, scenario diversity increases, and teams can stress-test autonomy stacks in a broader set of geographies without bespoke manual map construction.

The technical challenge is end-to-end and multi-stage. First, one must robustly identify drivable road surfaces under diverse appearance conditions (occlusions by trees/buildings, shadows, varying materials, seasonal effects, sensor noise). This is usually framed as semantic segmentation: a model predicts a dense road mask over pixels. Second, segmentation masks must be transformed into a vector representation—a graph with vertices (e.g., intersections, endpoints, turning points) and edges (road segments) that captures topology and connectivity. This “mask-to-graph” step is non-trivial because small segmentation defects (gaps, spurious blobs, boundary ambiguity) can cascade into broken connectivity and incorrect junction structure. Third, the graph must be geometrically regularized into a form compatible with simulator standards: reference lines need to be smoothed, lanes need to be parameterized, and intersections need explicit connectivity tables. Finally, the output must be serialized into a target map format (OpenDRIVE .xodr, potentially coupled with lanelet-style HD map layers [4]) and validated for consistency.

Since 2020, research has accelerated from “segmentation first then heuristic vectorization” toward more topology-aware and graph-native modeling, driven largely by metrics that penalize connectivity errors. A key example is the SpaceNet road extraction setting, where the dominant evaluation

emphasizes graph-theoretic similarity via Average Path Length Similarity (APLS), rather than purely pixel-wise overlap [5]. This aligns closely with downstream autonomy needs: a map with visually accurate roads, but broken connectivity is operationally suboptimal for route planning and simulation.

Topological consistency is critical not only for visual rendering but also for numerical investigation of traffic modes. As noted in recent studies on critical traffic modes, loss of connectivity can result from both flow dynamics and structural inconsistencies in the road graph [6].

While our pipeline focuses on the efficient generation of road networks, the downstream application of these maps often involves large-scale simulation campaigns. Such campaigns frequently rely on cluster computing systems to manage multiple independent tasks and ensure high reliability during extensive testing [7,8].

Comprehensive simulation environments for autonomous driving increasingly rely on multimodal data to ensure realism. While recent works have addressed the auditory dimension, proposing neural network algorithms for classifying traffic events based on audio patterns [9,10], our pipeline complements these efforts by automating the generation of the visual and geometric foundation.

High-fidelity digital twins also require accurate modeling of surface conditions. Recent approaches have demonstrated the feasibility of using acoustic sensors and machine learning to detect and classify pavement defects such as cracks and potholes [11]. Although our current work extracts layout from overhead imagery, the modular OpenDRIVE output is designed to accommodate additional attribute layers. This creates a pathway for integrating surface quality data - potentially sourced from methods like [11] directly into the simulator-ready map, enabling more realistic vehicle dynamics and maintenance scenario testing

In addition, the emergence of large “foundation” vision models has introduced new opportunities for domain transfer in remote sensing. Segment Anything (SAM) is a prominent promptable segmentation model trained at massive scale; it demonstrates strong generalization and can be adapted via fine-tuning for new visual domains [12]. For road mapping, this is particularly relevant because road appearance varies widely across regions, sensors, and seasons. Rather than training a bespoke segmentation network from scratch for each domain, a SAM-based approach suggests a more scalable operating model: leverage a strong pre-trained backbone, then specialize it for road geometry and intersection cues while retaining broad generalization.

This paper targets a pragmatic, simulator-focused roadmap: we aim to design a lightweight pipeline that (i) uses a SAM style segmenter to obtain high-quality road and intersection likelihood maps, (ii) converts these into a vector road graph with explicit junction topology, and (iii) compiles the resulting graph into an OpenDRIVE map suitable for CARLA ingestion. The overall objective is to reduce manual map authoring cost while providing a repeatable, extensible workflow that can scale to new regions with minimal incremental engineering.

Concretely, we make the following contributions:

- We consolidate recent progress in road segmentation and road graph extraction from overhead imagery, with a focus on topology correctness and downstream map compilation requirements.
- We propose a practical SAM-centered design pattern for road mask generation and intersection cue extraction, motivated by the transfer and generalization properties of foundation segmentation models [12].
- We outline a mask-to-graph and graph-to-OpenDRIVE compilation strategy that emphasizes consistency constraints (connectivity, smooth geometry, junction definitions), informed by known validation pitfalls in parametric OpenDRIVE road models [3].

II. RELATED WORK

We structure related work along the main stages of a simulator-oriented pipeline: (1) road segmentation from overhead imagery, (2) vectorization and topology recovery (mask to-graph), (3) graph-native road network extraction, and (4) compilation into simulator-compatible map formats such as OpenDRIVE and integration considerations for CARLA.

A. Road segmentation from satellite and aerial imagery

Road extraction has long been approached as a semantic segmentation task, where the output is a binary (or multi-class) mask over pixels. Encoder-decoder architectures such as U-Net [13] established an effective template for dense prediction by combining multi-scale context aggregation with spatially precise decoding. Subsequent segmentation families, including DeepLabv3+ [14], improved boundary quality and multi-scale reasoning through atrous spatial pyramid pooling and explicit decoders. Although these models were originally developed for natural image benchmarks, they have been widely adopted and adapted in remote sensing pipelines due to their strong baseline performance and straightforward training dynamics.

Within the 2020+ literature, a recurring theme is that roads are thin, elongated, and frequently occluded; therefore, segmentation models often need enhanced context reasoning to avoid fragmentation. SPIN Road Mapper proposes graph reasoning modules over spatial and interaction spaces to improve delineation under occlusions and clutter, and targets autonomous navigation relevance explicitly [15]. While such models improve pixel-level road detection, they still produce raster masks; topology correctness is not guaranteed, and downstream vectorization remains necessary.

Recent work also explores architecture modifications to pre-serve fine-grained connectivity cues. For example, UBR-Net introduces multi-scale attention and cross-residual encoding aimed at reducing disconnected predictions in challenging high-resolution scenes [16]. These approaches strengthen segmentation quality, but a persistent limitation remains segmentation metrics such as IoU or F1-score do not reliably correlate with route-level usability, especially near intersections and under overpasses. This is one reason graph-based evaluation (e.g., APLS) and graph-native modeling have gained momentum [5].

B. From raster masks to vector road graphs: post-processing and topology recovery

Converting a road mask into a usable road network graph is a critical but failure-prone stage. A typical classical stack involves skeletonization/thinning to obtain centerlines, junction detection by analyzing skeleton branch points, polyline simplification, and graph cleaning (spur removal, snapping, gap bridging). In practice, such heuristic pipelines are sensitive to small segmentation artifacts: a narrow occlusion-induced gap can sever connectivity; a small false-positive blob can create spurious intersections. These issues directly impact topology-driven metrics and downstream tasks like routing.

The modern literature increasingly treats topology recovery as a first-class objective rather than an afterthought. Sat2Graph introduces a graph-tensor encoding that allows learning to predict graph structure features in a dense, supervised manner, combining segmentation-style inference with explicit graph reconstruction [17]. A notable aspect is that Sat2Graph reports improvements on topology-aware metrics (including APLS) and discusses handling of non-planar structures (e.g., over passes), which are particularly relevant for simulation realism [17]. Still, even when a road graph is recovered, additional compilation is required to meet simulator format constraints (smooth parametric reference lines, explicit lane connectivity, and junction constructs).

C. Graph-native road network extraction

A second family of methods aims to predict road graphs more directly, reducing reliance on brittle heuristics. VecRoad frames road graph extraction as iterative point-based exploration that balances connectivity and alignment accuracy, and reports results in terms of graph quality rather than solely pixel overlap [18]. These iterative strategies tend to build graphs vertex-by-vertex or segment-by-segment, which can improve connectivity but can also introduce higher computational cost and complex control logic.

Single-Shot End-to-End Road Graph Extraction takes a more streamlined approach by combining a fully convolutional network that localizes points of interest (intersections, turns, endpoints) with a graph neural network that predicts links between points in a single pass [19]. This design reduces the operational overhead of iterative exploration and explicitly targets end-to-end trainability, which is attractive for productionizing a pipeline. However, the resulting graph still needs to be regularized and enriched to become a simulator-ready map (e.g., lane semantics, intersection geometry, and traffic rule elements).

Transformer-based methods have also been applied to road graph detection. RNGDet formulates graph generation as a sequential decision process with transformer-based perception and imitation learning, generating graphs vertex-by-vertex while emphasizing handling of complex intersections [20]. RNGDet++ enhances this line with additional supervision (instance segmentation) and multi-scale feature utilization, improving performance across public benchmarks [21]. These approaches are directly aligned with downstream autonomy applications (route planning, navigation priors), but their outputs typically remain at the level of a centerline graph rather

than a full OpenDRIVE representation, and they can still require non-trivial post-processing to ensure geometric smoothness and standard compliance.

D. Foundation segmentation models and SAM-based road mapping

Foundation models have reshaped the segmentation landscape by enabling promptable, transferable segmentation capabilities. Segment Anything (SAM) proposes a large scale, prompt-driven segmentation framework and demonstrates strong generalization across tasks and domains [12]. For remote sensing, the key promise is reduced domain adaptation cost: instead of training specialized models per geography/sensor, a foundation backbone can be adapted with less data and improved robustness.

A concrete example in road graph extraction is SAM-Road, which adapts SAM for large-scale vectorized road network graph extraction. SAM-Road fine-tunes the SAM image encoder to produce probability maps for roads and intersections, then extracts vertices via non-maximum suppression and predicts topology with a lightweight transformer-based graph network using SAM embeddings [22]. This is particularly relevant to simulator pipelines for two reasons. First, the method operationalizes a clean separation between geometry (mask-driven vertex placement) and topology (learned edge inference), which maps naturally onto a compilation stage. Second, the reported efficiency gains (fast inference over multi-kilometer regions) address scalability constraints that matter in simulation content generation.

Nevertheless, even SAM-centric graph extraction methods typically stop at producing a road graph, not a simulator compatible lane-level map. The gap between “graph output” and “OpenDRIVE-compliant road network with junctions and lane connectivity” remains substantial and underexplored as an end-to-end research focus.

E. Simulator-compatible map generation and OpenDRIVE constraints

CARLA provides an open simulation stack that is widely used for autonomy research [1], and OpenDRIVE is a widely supported standard for exchanging parametric road networks [2]. From a systems integration viewpoint, generating OpenDRIVE from extracted graphs requires more than correct connectivity: the standard expects a consistent geometric parameterization (reference lines, lane offsets/widths), explicit junction definitions, and continuity properties (e.g., smooth curvature) that can be violated by noisy or piecewise-linear graph estimates.

Empirical validation work demonstrates that even when a road network is conceptually correct, parametric inconsistencies can cause downstream simulators to exhibit artifacts (gaps, overlaps, invalid curvature transitions) or outright reject maps [3]. Therefore, “graph-to-OpenDRIVE compilation” should be treated as a constrained optimization and validation problem, not merely a file format conversion. In practical pipelines, this often entails (i) smoothing and fitting parametric curves to centerlines, (ii) generating lane boundaries and lane sections, (iii) deriving junction polygons and lane-to-

lane connectivity, and (iv) running validators to enforce standard constraints before deployment in CARLA-like runtimes.

F. Summary of gaps

Across the post-2020 literature, methods have substantially improved road segmentation and road graph extraction quality, especially under topology-aware evaluation. However, several gaps remain when the end goal is simulation-ready maps:

- Standard-compliant map compilation is rarely end-to-end addressed. Most papers terminate at masks or centerline graphs, leaving OpenDRIVE generation to ad hoc engineering.
- Junction topology vs. junction geometry remains a weak link. Graph connectivity may be correct while geometric layout (turn radii, lane fan-in/fan-out, continuity) is not sufficient for simulator realism.
- Lane-level semantics are under-specified. Overhead imagery alone often does not uniquely determine lane count, lane directions, and traffic rules, yet these are required for high-fidelity simulation.
- Validation and robustness are not standardized. OpenDRIVE correctness constraints and simulator ingestion failure modes are not commonly integrated into learning objectives or reported metrics, despite evidence that validation is critical [3].

These gaps motivate pipelines that explicitly couple modern segmentation/graph extraction (including SAM-based components) with compilation and validation layers targeting OpenDRIVE and CARLA integration.

III. PROBLEM STATEMENT AND FORMULATION

We study simulator-ready road map generation from overhead imagery. Given an orthorectified satellite/aerial RGB image I , our goal is to automatically produce an ASAM OpenDRIVE (.xodr) map O that can be loaded in CARLA [1] and is structurally consistent with the OpenDRIVE data model (roads, lanes, and junction connectivity) [2]. Unlike prior work that stops at raster masks or centerline polylines, we explicitly target a complete xodr artifact usable by an autonomy simulator.

A. Inputs, output, and success criteria

Input: an image $I: \Omega \rightarrow \mathbb{R}^3$, $\Omega \subset \mathbb{Z}^2$, optionally with a geo-transform $T: \Omega \rightarrow \mathbb{R}^2$ (pixel $\rightarrow$ metric ground plane). We also allow optional prompts π when using a promptable segmenter.

Output: an OpenDRIVE map $O \in \mathcal{O}$ (serialized as .xodr) describing (i) road reference lines, (ii) lane sections attached to each road, and (iii) junctions with valid lane connectivity [2].

Definition of Done:

- 1) *Ingestion*: O is successfully imported in CARLA [1].
- 2) *Connectivity*: the derived road network is sufficiently connected for routing (no systematic breaks in the main component).

3) *Junction topology*: intersections are represented as junction entities with valid connections (incoming/outgoing roads and lane links) [2].

B. Compact formulation

We express the pipeline as a composition:

$$I \xrightarrow{f_\theta} P \xrightarrow{\tau} M \xrightarrow{\Psi} G \xrightarrow{\Phi} O$$

Here f_θ is a (SAM-based) segmenter producing probability maps P for roads (and optionally junction likelihood), M is a cleaned binary mask obtained by thresholding with τ , Ψ converts the mask into a vector road graph $G = (V, E)$, and Φ compiles G into an OpenDRIVE map. We use SAM as a promptable segmentation backbone due to its transfer capability [12] and note that SAM-Road demonstrates feasibility of extracting road vertices and topology using SAM embeddings plus a lightweight graph module [22].

a) Mask-to-graph Ψ (minimal): From the road mask M , we compute a centerline skeleton $S = \text{Skel}(M)$. We extract graph vertices as endpoints and branch points on S , trace centerline chains between vertices to form edges, and apply stabilization (snapping nearby vertices, pruning short spurs, and bridging small gaps when headings are compatible). If a geo-transform T is available, vertex coordinates are mapped to R2 directly; otherwise, we work in pixel space with a global scale.

b) Graph-to-OpenDRIVE compilation Φ : Each graph edge is converted into a smooth road reference line (e.g., spline/clothoid fit), then expanded into lane sections using a minimal parameter set (default lane count and lane width). Junction nodes are compiled into OpenDRIVE junction objects by generating connecting roads between feasible incoming/outgoing directions and defining lane links, following OpenDRIVE connectivity requirements [2]. Since OpenDRIVE validity goes beyond XML schema checks, we treat $\text{Valid}(O) = 1$ as an explicit requirement, following known validation pitfalls [3].

c) Evaluation: When a reference graph G^* is available (e.g., SpaceNet-style-labels), we evaluate topology with APLS, which emphasizes navigable connectivity rather than pixel overlap [5, 23]. In the no-GT setting, we report proxy checks (component size, number of disconnected endpoints, and simulator ingestion success).

IV. DISCUSSION

A. Preparation of experimental data

The proposed pipeline demonstrates that a simulator-ready road map can be generated from overhead imagery using a small number of well-separated components, with the Segment Anything Model (SAM) acting as the core perception module. In contrast to fully end-to-end road graph prediction systems, our approach emphasizes modularity and controllability: segmentation, topology reconstruction, and OpenDRIVE compilation are treated as distinct stages. This separation

simplifies error analysis and makes the pipeline easier to adapt to new regions or simulator constraints.

A central empirical observation is that SAM can reliably identify road surfaces across diverse visual conditions without being explicitly trained for road segmentation. Even in a zero shot or weakly prompted setting, SAM produces coherent masks for elongated, low-texture structures that correspond to drivable roads. This supports the hypothesis that foundation segmentation models can substantially reduce the effort required to bootstrap large-scale road extraction pipelines.

At the same time, the figure highlights an important limitation: SAM optimizes object-level segmentation rather than topological correctness. As a result, the raw masks may contain small gaps, merged regions at intersections, or ambiguous boundaries between adjacent roads. While such imperfections are acceptable in many vision tasks, they can lead to connectivity errors after skeletonization and graph extraction. Our experiments confirm that lightweight geometric post-processing is essential to compensate for these effects and to recover a usable road topology.

Algorithm 1 SAM-to-OpenDRIVE Road Map Generation

Require: Overhead image I ; optional geo-transform T ; optional prompts π ; thresholds $\tau, \varepsilon, \ell_{\min}$

Ensure: OpenDRIVE map O (.xodr)

- 1: **Segmentation:** compute road (and optionally junction) probabilities $P \leftarrow f_{\theta}(I, \pi)$ [12]
 - 2: **Binarize & clean:** $M \leftarrow \mathbf{1}[P_{\text{road}} \geq \tau]$; apply morphological cleanup
 - 3: **Skeletonize:** $S \leftarrow \text{Skel}(M)$
 - 4: **Graph extraction:** detect vertices V (endpoints/branchpoints) on S ; trace chains to form edges E
 - 5: **Graph stabilization:** snap vertices within ε ; prune spurs shorter than $\ell_{\min}$; bridge small gaps
 - 6: **Metric coordinates:** if T exists, map $V \leftarrow T(V)$ and polylines $E \leftarrow T(E)$
 - 7: **Geometry fitting:** fit smooth reference lines to each edge polyline (spline/clothoid)
 - 8: **Lanes:** assign minimal lane model (default lane count/width) and generate lane sections [2]
 - 9: **Junctions:** for each v with $\deg(v) \geq 3$, create an OpenDRIVE junction with connecting roads and lane links [2]
 - 10: **Export:** serialize to OpenDRIVE file O
 - 11: **Validate:** run structural and geometric checks; fix violations when possible [3]
 - 12: **return** O
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Fig. 1. Example of road surface segmentation produced by the Segment Anything Model (SAM).

Another key aspect discussed in this work is the gap between vector road graphs and simulator-compatible representations. Even when a road graph is topologically reasonable, additional constraints imposed by the OpenDRIVE standard—such as smooth reference lines, consistent lane connectivity, and valid junction definitions—must be satisfied for successful simulator ingestion. Treating OpenDRIVE validity as an explicit requirement, rather than an afterthought, proved crucial for robust integration with CARLA.

Overall, the discussion indicates that SAM-based pipelines are particularly well suited for rapid map prototyping and large-scale simulation content generation. Their main strength lies in scalability and ease of deployment, while their main limitations concern fine-grained semantics and complex multi-level road structures.

V. CONCLUSION

This paper presented a lightweight and modular pipeline for generating simulator-ready road maps from overhead imagery using the Segment Anything Model as the primary segmentation component. Starting from a single satellite or aerial image, the proposed approach extracts road masks, reconstructs a vector road graph, and compiles it into an ASAM OpenDRIVE map that can be directly loaded into the CARLA simulator.

The main contribution of this work is the demonstration that modern foundation segmentation models can significantly simplify the perception stage of map generation without sacrificing practical usability. By combining SAM-based segmentation with classical geometric and graph-processing techniques, we bridge the gap between image-based road detection and simulator-compatible map formats. Unlike many existing approaches, our pipeline explicitly targets OpenDRIVE compliance and simulator ingestion as success criteria.

While the current implementation focuses on simplicity and robustness, it naturally opens several directions for future work. These include incorporating learned topology

refinement, inferring lane-level attributes and traffic semantics, and extending the pipeline to multi-modal or temporal data sources. We believe that such extensions can further close the gap between automated map generation and high-fidelity HD maps, while preserving the scalability benefits demonstrated in this study.

In summary, the proposed SAM-to-OpenDRIVE pipeline provides a practical baseline for rapid generation of diverse simulation environments and illustrates the potential of foundation models as key enablers for next-generation map creation workflows in autonomous driving research.

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